



Design and field trial of EmotionFrame: exploring self-journaling experiences in homes for archiving personal feelings about daily events

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ABSTRACT

Capturing people's spontaneous thoughts or daily emotions in their everyday living space can be difficult and can pose a psychological burden from a clinical standpoint. To address this issue, we have developed EmotionFrame, which allows users to archive their daily experiences by recording videos of their current thoughts and emotions at home. EmotionFrame serves as a journaling device, and the parameters for journaling have been determined based on a depression assessment model. This study explores the benefits of users' recording feelings about their daily events, which can also help with assessing depressive mood with the journaling data. Our three-week in-field study with eight participants showed that EmotionFrame could be integrated into participants' daily routines based on their previous diary-writing habits. In addition, its harmonious frame shape fits in with the everyday environment. Being able to engage in spontaneous journaling in private spaces – and in a way that allowed them to see themselves – prompted the study participants to effectively record their sincere thoughts and emotions during the archiving process.

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1. Introduction

Digital self-journaling allows users to collect vast amounts of information about themselves by self-recording their daily experiences, activities, health, and emotions. Accumulating self-relevant information helps the users to gain knowledge about themselves (Rapp & Cena, 2016). In addition, self-understanding from personal informatics could be used for clinical purposes (Chung et al., 2016; Elliott et al., 1996; Kelley et al., 2017; Yamashita et al., 2017), as it could inform the users of clinically relevant information and trigger behavioral change. Moreover, manual self-recording, as opposed to automatic recording, can allow one to organize information from a personal point of view and to think about one's internal state (Ayobi et al., 2020; Y.-H. Kim et al., 2017). In other words, it is suggested that the user can recognize the records collected through the manual record as their own, so they can easily identify with the records and attach meaning (Sengers & Gaver, 2006) to them.

Self-journaling can affect how users look back on themselves depending on how they recorded their events. In the HCI research stream, self-journaling research using handwriting and voice

recording has actively been conducted. Lindström's research team designed the Affective Diary (Elsden et al., 2016; Lindström et al., 2006), which automatically collects the user's bodily memorabilia and media on mobile devices, then provides the collected information when the user writes a diary with a digitized pen. The researchers found that users could interpret, understand, and record their motions and media through the Affective Diary. Mols's research team (Mols et al., 2020) utilized tangible interfaces for reflection through user-created media (e.g., text, audio) through the pyramid-shaped Cogito to annotate text (via handwriting) and the scale-shaped Balance to capture voice notes. Moreover, Sawhney et al. (2018) mentioned that audio journaling is an unobstructed and suitable method for capturing spontaneous and introspective experiences with a focus on emotional resonance.

Presenting a life narrative can be beneficial for one's mental health by externalizing issues related to life events through writing or speaking, as reported by (Wallis et al., 2011). De Vecchi et al. (2016) conducted review research which confirmed that there is potential for mental health benefits that arise from understanding and emphasizing daily experiences. One such benefit of presenting a life narrative is *mindfulness*, which involves accepting emotions derived from personal experiences. This could potentially improve emotional well-being (Kabat-Zinn, 2003) by reducing concern toward uncontrollable stimuli, such as emotions and experiences (Wiech et al., 2006).

In addition, research on video journaling has been conducted. Regarding health-related behavioral change, one study (Melton et al., 2013) compared video journaling to other journaling methods (i.e., pen and pencil). They found that the video-journaling group was more satisfied with the action of journaling as well as journaling maintenance. In addition, Parikh et al. (2012) conducted a study in which they collected students' internship experiences through video journaling and found that this method led to an increase in the students' confidence and genuine reflection. Clarke (2009) also found that the video journaling allows for fast and honest expression of complex emotions encountered during fieldwork.

Furthermore, video journaling in the form of vlogging using video or social network platforms (e.g., YouTube, Instagram) has also been used recently. Griffith and Papacharissi (2010) remarked that daily records through vlogging can be used as a means of self-expression. Similarly, Brott and Willis (2021) confirmed that vlogging is a way of providing meaning to one's experiences and allowing self-reflection.

The above-mentioned studies suggest that various means can be used for self-journaling, and a user may feel different emotions and experiences depending on his or her recording method. In addition, attempts to evaluate the user's state (e.g., mental state, depressive state) through the experience sampling method (ESM), especially based on the user's diary (one of the self-journaling methods), have been recently made. Berkel et al. (2017) confirmed that this approach is a reliable method for accurately recalling and recording the relatively close past by allowing users to archive their daily lives. In addition, it is reliable enough to be used for medical purposes (Csikszentmihalyi & Larson, 2014). In addition, Christoff et al. (2016) and Andrews-Hanna et al. (2013) suggested that user-generated spontaneous thought and the user's evaluation of those thoughts (e.g., valence, self-relevance, etc.) could be used to assess the user's depressive mood. Furthermore, accumulated recordings of self-journaling can provide information on spontaneous thought dynamics, which can also be used to assess one's mood state (Christoff et al., 2016).

To sum up, recording a diary about everyday experiences enables self-reflection and self-understanding. At the same time, data about self-generated spontaneous thoughts and their dynamics could be used to assess depressive mood. This suggests a possibility of evaluating depression through storing and retrieving users' recordings in their daily lives. However, it is not easy to maintain such records. Thus, new design interventions are needed to help the user to conduct self-journaling without lapsing, or stopping the process of recording about himself or herself. Epstein et al. (2015) discovered four factors (i.e., forgetting, upkeep, skipping, and suspending) that caused users to stop self-tracking. In addition, many researchers (Intille et al., 2016; Paruthi et al., 2018; van Berkel et al., 2022) mentioned that the length of time required to access a tracking app

after one decides to record could act as a barrier to recording. They utilized a tangible device to reduce the time required for daily recording.

To reduce these barriers and naturally prompt self-reflection among users by encouraging them to record daily events and feelings as part of their everyday routine, previous studies have suggested using appropriate ambient reminders (including physical presence (Mols et al., 2020; Paruthi et al., 2018) or ambient notification (Baharin et al., 2015; Gustafsson & Gyllenswärd, 2005; Müller et al., 2013; Papachristos et al., 2020) for triggering their journaling. In this regard, researchers (K. Lee & Hong, 2017; Sarzotti, 2018) have actively studied self-recording via the tangible form of a data – collection device. Especially, Mols et al. (2020) confirmed that the device's physical presence could remain eye-catching and help the user better record their data or access the data contained in the interfaces. Similarly, Paruthi et al. (2018) revealed that a tangible self-report device could help users to keep recording through the physical presence itself. In addition, a proper intervention degree that can be applied in users' environments is essential for deep integration into their daily lives. Regarding this, previous works have provided sounds (Baharin et al., 2015; Butz & Jung, 2005) and light (Gustafsson & Gyllenswärd, 2005; Müller et al.; Papachristos et al., 2020) for ambient awareness of TUI. In addition, aesthetic features should also be considered to integrate an interactive device into a daily routine (Park et al., 2019); and further studies (Mols et al., 2020)

This strand of previous research has opened a new design space for (1) designing a self-journaling device collecting users' daily events and self-evaluations, which could also be used for assessing depressive mood, and (2) an appropriate system of intervention for prompting recording in an everyday living space.

However, before users' depressive moods could be assessed through the daily self-journaling data collected from devices in everyday living spaces, the following issues need to be addressed: (1) appropriate device design and journaling parameters for promoting recording in everyday life; (2) examining in-situ experiences of collecting thoughts on life events through the self-journaling device, which can aid in the assessment of depressive mood in daily life; and (3) investigating whether the spontaneous thoughts and content variables contained in the collected records were suitable for applying Andrews-Hanna et al. (2013) method for the assessment of depression.

Accordingly, the scope of this study is to investigate the design and implementation of a self-journaling device based on a depression assessment model adopted from a previous study (Andrews-Hanna et al., 2013; B. Kim et al., 2022) and to examine how users accept and adopt their daily recordings through this device before we actually use the device to assess depression. In particular, we focused on the in-situ experience of using the EmotionFrame and its acceptability, rather than on reflection or interpretation of past records as a clinical diagnostic tool.

To this end, we pursued the following research questions:

RQ-1) How might a self-journaling device based on a depression assessment model be designed?

- (1) To answer this question, we employed research through a design approach (Zimmerman et al., 2007) and developed an interactive frame-shaped device, called EmotionFrame (Figure 1). It is a self-journaling device that allows users to archive their daily experiences at home through video recording and a simple questionnaire to assess emotional states adopted from Andrews-Hanna et al. (2013)

RQ 2) How can users accept and adopt their experience of capturing their everyday thoughts and emotions through the EmotionFrame device in their homes?